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Lab Report: Antenna Characterization

Experiment performed: Part B: Effect of Distance, Part C: Co-Polarization Beam Pattern, Part D: Cross-Polarization Beam Pattern, and Part E: Half-Power Beamwidth and Directivity.

Note: Part A (VSWR and antenna operating-frequency measurement) was not performed this session; the lab instructor deferred it to a later session.

1. Objective

- To measure the effect of distance on the received power from the antenna and examine the observed trend against the free-space inverse-square law.
- To estimate the antenna's approximate operating frequency from a manual measurement of its physical (monopole) length, since the VSWR-based determination (Part A) was not performed.
- To measure the far-field power pattern of the antenna with respect to angle, in both co-polarization and cross-polarization configurations, for the E-plane and H-plane cuts.
- To determine the half-power beamwidth (HPBW) in the E-plane and H-plane from the co-polarized patterns, and to estimate the antenna's directivity.

2. Theory

2.1 Antenna Basics

An antenna is a transducer that converts electrical power into electromagnetic waves and vice versa. A dipole antenna consists of two equal metal arms fed at the centre, with total length approximately λ/2. A monopole antenna is approximately half of a dipole mounted above a conducting ground plane, with length approximately λ/4; the ground plane acts as a mirror, so the monopole behaves like a full dipole of twice its physical length.

For a quarter-wave monopole operating at its fundamental resonance, the physical length L is related to the free-space wavelength λ by:

$L = \lambda / 4 \qquad \Rightarrow \lambda = 4L,\ f = c / \lambda$

where $c \approx 3 \times 10^8$ m/s is the speed of light in vacuum.

2.2 Antenna Beam Pattern

The radiation pattern of an antenna generally consists of a main lobe (the direction of maximum radiated energy, indicating the principal beam direction), one or more side lobes (radiation distributed sideways), and a back lobe (opposite to the main lobe).

The E-plane is the plane containing the electric-field vector and the direction of propagation, while the H-plane is the plane containing the magnetic-field vector and the direction of propagation. The E-plane and H-plane are perpendicular to each other.

2.3 Co-Polarization and Cross-Polarization

Co-polarization refers to a measurement configuration in which the transmitting and receiving antennas' electric-field vectors are aligned (parallel), so that the receiver couples maximally to the transmitted wave. Cross-polarization refers to a configuration in which the two E-field vectors are orthogonal (perpendicular); ideally the received power should then be very low, and in practice the residual received power (never exactly zero, due to imperfect antenna polarization purity and multipath) is used to quantify the cross-polarization discrimination of the antenna system.

2.4 Half-Power Beamwidth

The half-power beamwidth (HPBW) is the angular width of the main lobe between the two points where the received power drops to half of its maximum value. A small HPBW indicates a narrow, focused beam; a large HPBW indicates a broad beam.

2.5 Directivity

Directivity measures how concentrated the radiated power is in the direction of the main lobe compared to an isotropic radiator. For practical antennas it can be approximated from the E-plane and H-plane half-power beamwidths (θ_E, θ_H, expressed in radians) as:

$D \approx 4\pi / (\theta_E \cdot \theta_H)$

3. Components Required

- Dipole (transmitting) and horn (receiving) antennas
- RF signal generator
- Spectrum analyser
- Rotatable antenna support stand (“chair”), 0–360°
- Measuring tape (for distance and physical antenna-length measurement)
- Coaxial cables and suitable RF adapters

(The oscilloscope, RF Tee connector, and 50 Ω termination used for the VSWR measurement in Part A were not required, as that part was not performed in this session.)

4. Experimental Setup

The transmitting dipole antenna was mounted on a tripod and connected to the RF signal generator using a cable. The receiving horn antenna was connected to the spectrum

analyser. This followed the general arrangement of Fig. 4 of the lab manual, with the transmitter and receiver separated by a distance D and aligned at the same height

For the beam-pattern measurements, the receiving horn was mounted on a rotatable stand ("chair") that could be rotated through a full 0–360° in a single azimuthal sweep. Rather than using two independent mechanical rotations of the DUT to trace the E-plane and H-plane cuts separately (as suggested in Figs. 6–7 of the manual), a single rotation axis was used throughout, and the plane being cut (E-plane or H-plane) was instead selected by re-orienting the polarization (tilt) of the transmitting dipole and/or the receiving horn by 90° between measurement runs. This is an equivalent alternative technique for tracing the two principal-plane cuts and is described further in Section 9 (Learnings).

5. Experimental Procedure

5.1 Estimating the Antenna Operating Frequency (Part A not performed)

Since Part A (VSWR sweep) was deferred by the lab instructor to a later session, the antenna's approximate operating frequency was instead estimated as follows:

- The physical length of the monopole element of the horn antenna was measured manually with a ruler/measuring tape as L ≈ 5 cm (this measurement carries appreciable uncertainty, estimated at ± 0.2–0.5 cm, due to the difficulty of precisely locating the feed point and tip of the element by hand).
- Treating the element as a quarter-wave monopole (L = λ/4), the free-space wavelength and hence the operating frequency were estimated (see Section 7.1 for the calculation).

5.2 Part B: Effect of Distance

- The transmitting and receiving antennas were aligned with the same polarization and kept at the same height.
- The signal generator frequency was set to 1.3 GHz with Pin = 0 dBm.
- An unrelated horn antenna belonging to another experiment, located just outside the lab about 1–2 m to the side of our setup, was found to be interfering with the measurement: with it in place, the received power became fainter as the receiving antenna was brought closer to the transmitter, the opposite of the expected trend, giving highly ambiguous readings. The interfering antenna was therefore moved further away (≈ 3 m) before proceeding; see Section 6.2 for further discussion.
- The distance R between the antennas was varied, and the received power P_r was recorded from the spectrum analyser at each distance.
- P_r was plotted against R and against 20 log10(R) to examine the received-power fall-off with distance.

5.3 Part C: Co-Polarization Beam Pattern

- The transmitting frequency was set to 1.3 GHz within the estimated operating range of the antenna.
- With the transmitter and receiver E-fields aligned (co-polarized), the receiving horn was rotated through 0–360° and the received power was recorded at 20° intervals; this sweep, with the dipole vertical, traced the H-plane cut (“Co-plane H”).
- The dipole was then tilted by 90° (with the horn re-aligned to remain co-polarized) and the same 0–360° sweep was repeated, this time tracing the E-plane cut (“Co-plane E”).
- The E-plane and H-plane co-polarized patterns were plotted as polar plots.

5.4 Part D: Cross-Polarization Beam Pattern

- Retaining the same frequency, antenna separation, height, and angular steps as the co-polarization measurement, the receiving horn (or dipole) was rotated by 90° relative to the transmitter so that the two E-fields were orthogonal (cross-polarized).
- The 0–360° sweep was repeated in this cross-polarized configuration for both the E-plane and H-plane cuts.
- The cross-polarized E-plane and H-plane patterns were plotted and compared against the corresponding co-polarized patterns.

5.5 Part E: Half-Power Beamwidth and Directivity

- From the co-polarized E-plane pattern, the two points about the main-lobe peak were identified and used to calculate θE.
- From the co-polarized H-plane pattern, the two points about the main-lobe peak were identified and used to calculate θH.
- θ_E and θ_H were converted to radians and used to estimate the directivity $D \approx 4\pi/(\theta E \theta H)$.

6. Observations and Graphs

6.1 Antenna Operating Frequency (from physical length)

Quantity	Value
Measured monopole length, L	≈ 5 cm (± 0.2–0.5 cm, manual measurement)
Wavelength, $\lambda = 4L$	20 cm
Estimated operating frequency, $f = c/\lambda$	≈ 1.5 GHz

6.2 Part B: Received Power vs Distance (f = 1.3 GHz, Pin = 0 dBm)

Initial trial (not tabulated): with an unrelated horn antenna from another experiment sitting ≈ 1–2 m to the side of our setup, the received power was observed to get fainter as the receiving antenna was moved closer to the transmitter, the opposite of the physically expected behaviour, making the readings unusable. This was traced to interference from that nearby antenna. Moving it about 3 m further away resolved the issue, and the consistent, physically sensible readings below were then recorded.

R (m)	Pin (dBm)	Pout (dBm)
8.12	0	-61.2
7.35	0	-54.7
6.58	0	-51.2

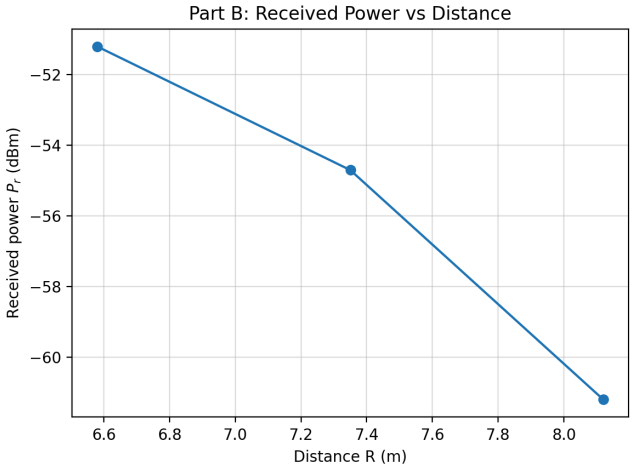


Figure 1: Received power Pr vs distance R

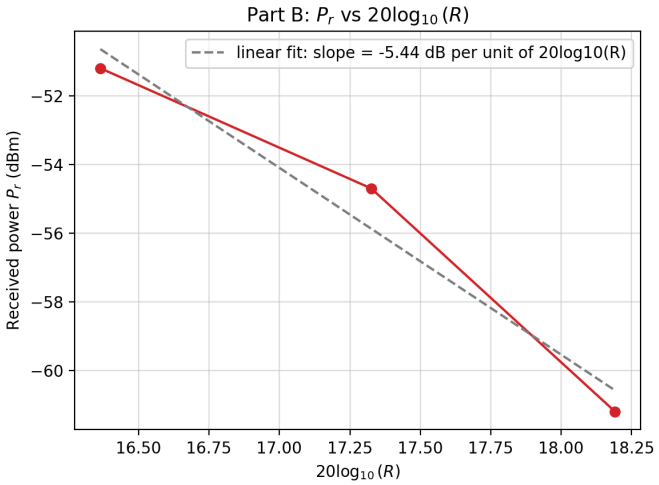


Figure 2: Received power Pr vs 20 log10(R)

6.3 Part C: Co-Polarization Beam Pattern

6.3.1 Co-plane, H-plane cut (f = 1.3 GHz, Pin = 0 dBm)

θ (deg)	Pout (dBm)
0	-58.0
20	-60.50
40	-66.90
60	-75.00
80	-72.00
100	-70.00
120	-73.90
140	-66.50
160	-65.20
180	-67.13
200	-64.50
220	-70.50
240	-74.70
260	-63.60
280	-61.50
300	-60.80
320	-68.30
340	-62.30
360	-58.80

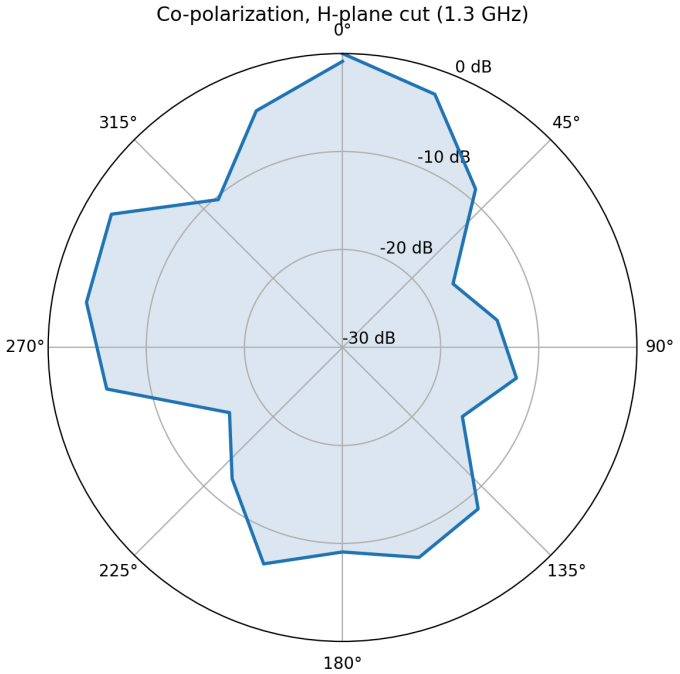


Figure 3: Polar plot – Co-polarization, H-plane cut

6.3.2 Co-plane, E-plane cut

θ (deg)	Pout (dBm)
0	-53.00
20	-54.10
40	-55.30
60	-56.40
80	-61.20
100	-64.10
120	-61.02
140	-70.40
160	-68.80
180	-61.70
200	-60.10
220	-70.30
240	-58.10
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300	-57.60
320	-62.40
340	-57.40
360	-53.40

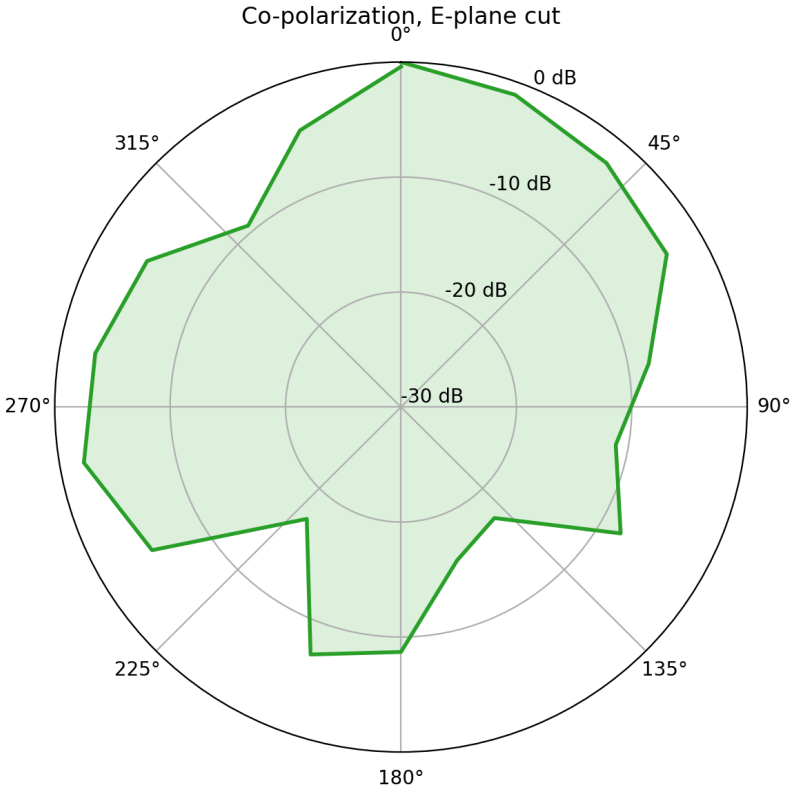


Figure 4: Polar plot – Co-polarization, E-plane cut

6.4 Part D: Cross-Polarization Beam Pattern

6.4.1 Cross-plane, E-plane cut (Pin = 0 dBm)

θ (deg)	Pout (dBm)
0	-64.71
20	-63.79
40	-64.67
60	-65.10
80	-71.34
100	-73.50
120	-74.68
140	-80.47
160	-75.21
180	-77.48
200	-81.37
220	-77.46
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260	-70.75
280	-69.52
300	-68.67
320	-64.21
340	-63.05
360	-64.16

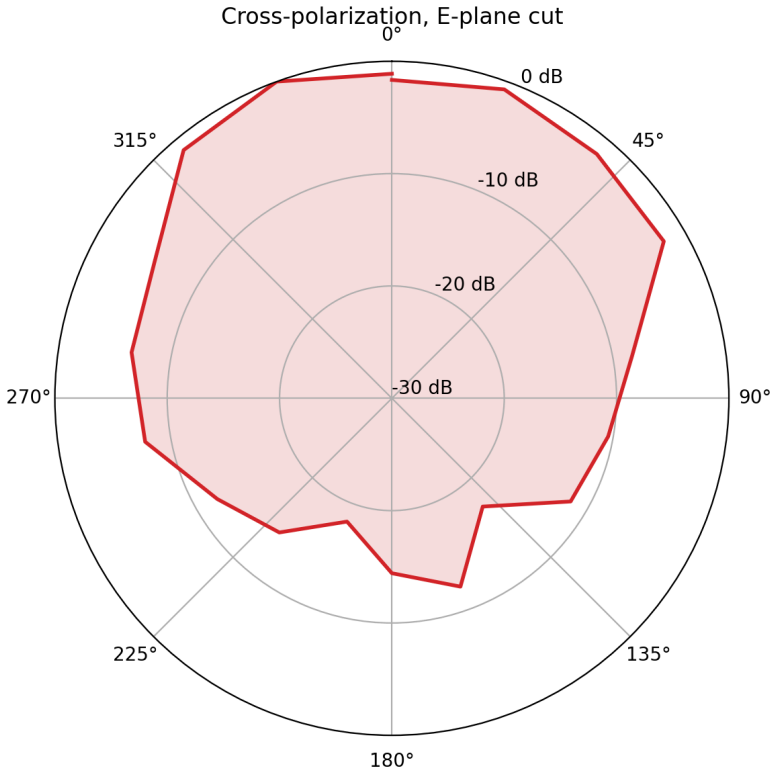


Figure 5: Polar plot – Cross-polarization, E-plane cut

6.4.2 Cross-plane, H-plane cut

θ (deg)	Pout (dBm)
0	-65.50
20	-76.50
40	-81.50
60	-70.60
80	-73.90
100	-71.00
120	-65.00
140	-73.20
160	-71.50
180	-70.25
200	-69.80
220	-69.20
240	-72.90
260	-64.20
280	-63.70
300	-61.10
320	-59.20
340	-60.20
360	-63.70

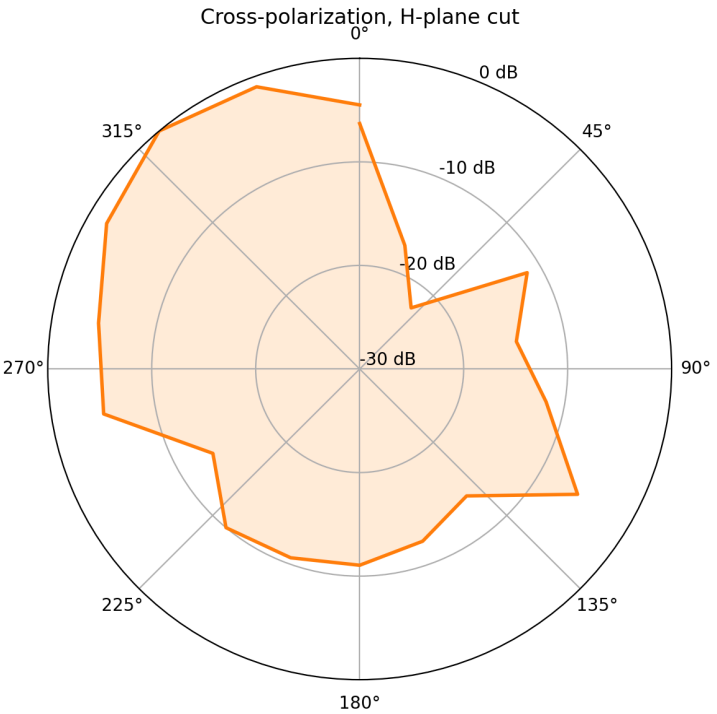


Figure 6: Polar plot – Cross-polarization, H-plane cut

6.5 Co- vs Cross-Polarization Comparison

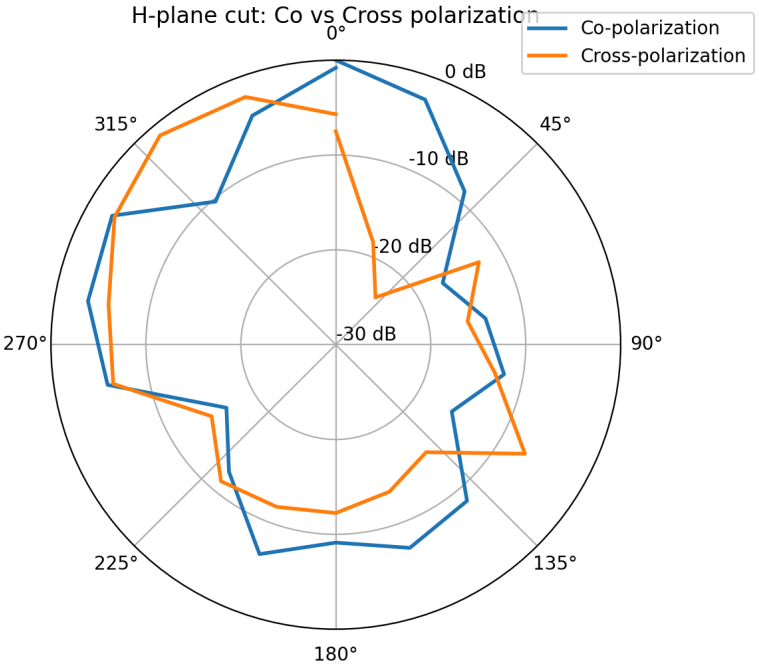


Figure 7: Overlay polar plot – H-plane cut, Co vs Cross polarization

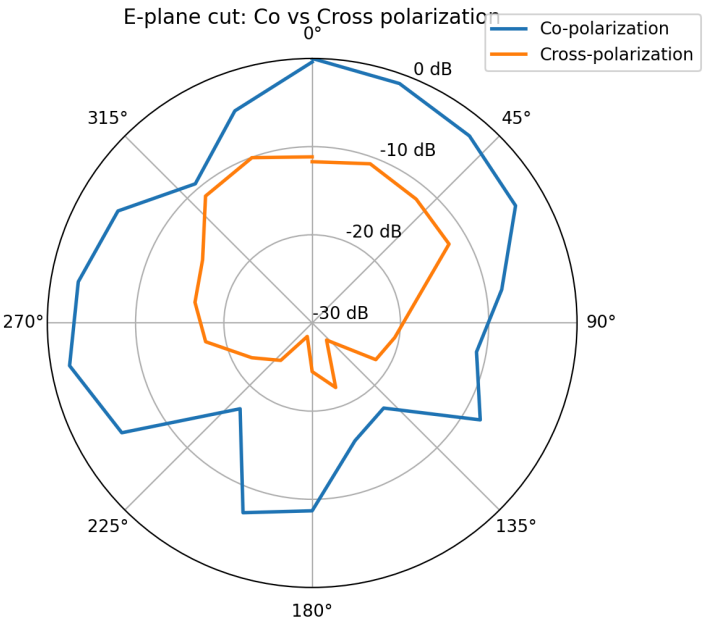


Figure 8: Overlay polar plot – E-plane cut, Co vs Cross polarization

7. Calculations and Results

7.1 Antenna Operating Frequency

Using L ≈ 5 cm and treating the element as a quarter-wave monopole:

$\lambda = 4L = 4 \times 5\ \text{cm} = 20\ \text{cm} = 0.20\ \text{m}$

$f = c / \lambda = (3 \times 10^8\ \text{m/s}) / (0.20\ \text{m}) = 1.5 \times 10^9\ \text{Hz} = 1.5\ \text{GHz}$

This estimate is reasonably close to the 1.3 GHz used for the subsequent measurements.

7.2 Part B: Received Power vs Distance

As noted in Section 6.2, an initial trial was corrupted by interference from an unrelated horn antenna ≈ 1–2 m to the side of the setup, which produced a physically impossible trend (power falling as the receiver approached the transmitter). After moving that antenna ≈ 3 m further away, the received power increased as the distance was decreased (from −61.2 dBm at 8.12 m to −51.2 dBm at 6.58 m), consistent with the expected inverse-square-law fall-off of received power with distance.

A linear fit of Pr against 20 log10(R) over the three measured points gives a slope of approximately −5.4 (dB per unit of 20 log10(R)). Even after relocating the interfering antenna, some residual stray coupling or multipath from it (and from other reflecting surfaces in the lab) may still have been present at the ≈ 1.5 m range of distances probed here; with only three data points, this cannot be separated from ordinary measurement scatter (see Section 10).

For an ideal free-space link, doubling the distance is expected to reduce the received power by 20 log10(2) ≈ 6.02 dB.

7.3 Part E: Half-Power Beamwidth and Directivity

The −3 dB threshold was located by identifying the main-lobe peak in each co-polarized pattern and interpolating linearly between adjacent measured points to find the two angles at which the power falls 3 dB below the peak.

Cut	Peak Pout (dBm)	Peak angle (deg)	−3 dB points (deg)	HPBW (deg)
Co-plane E	−53.00	0	−13.6, +52.7	66.4
Co-plane H	−58.80	0	−17.1, +24.1	41.2

Converting to radians: $\theta_E = 66.4° = 1.159$ rad, $\theta_H = 41.2° = 0.719$ rad.

$D \approx 4\pi / (\theta_E \cdot \theta_H) = 4\pi / (1.159 \times 0.719) \approx 15.1$

8. Results Table

Quantity	Symbol	Value
Estimated operating frequency (from monopole length)	f	≈ 1.5 GHz
E-plane half-power beamwidth	θ_E	66.4°
H-plane half-power beamwidth	θ_H	41.2°
Estimated directivity	D	≈ 15.1

9. Learnings

This section summarizes my understanding of the co-/cross-polarization and E-/H-plane concepts used throughout the experiment, along with how they applied to our specific setup.

- A linear antenna such as a dipole or monopole radiates an electric field that is polarized along the direction of the antenna's own physical axis; this is the direction along which the charges/current oscillate in the conductor. The plane containing this E-field vector and the direction of wave propagation is called the E-plane; the plane perpendicular to it (containing the magnetic-field vector and the propagation direction) is the H-plane. The E-plane and H-plane are always perpendicular to each other.
- Co-polarization occurs when the transmitting and receiving antennas' E-fields are aligned (parallel); in this configuration the receiver couples maximally to the incoming wave. Cross-polarization occurs when the two E-fields are perpendicular to each other; ideally, the received power should then be minimal, and the amount of power still received (rather than being exactly zero) is a practical measure of the antenna system's polarization purity.
- In our setup, the transmitter was a dipole and the receiver was a horn antenna mounted on a rotatable stand (“chair”) that sweeps through a single 0–360° azimuthal rotation. To still obtain the four required cuts, the plane being sampled was instead selected by re-orienting the polarizations of the transmitter/receiver by 90° between runs: with both E-fields vertical and parallel (co-polarized) and the chair swept through 360°, the sweep traced the H-plane cut (“Co-plane H”); turning the horn 90° (now cross-polarized relative to the still-vertical dipole) and repeating the same chair sweep gave the “cross-plane E” cut; tilting the dipole itself by 90° (E-field now horizontal, restoring co-polarization with the rotated horn) and sweeping the chair gave the “co-plane E” cut; and finally returning the horn to its original orientation (now cross-polarized again, with the dipole still tilted) and sweeping the chair gave the “cross-plane H” cut.
- The Part B readings showed that a set of measurements can look fine on its own while still being physically wrong if another RF source nearby is coupling into the receiver. Since power should rise as the receiver gets closer, seeing the opposite trend was the clue that something outside our setup was the problem, so it is worth checking for nearby RF equipment before trusting a power measurement.

10. Conclusions

- Since Part A was not performed, the antenna's operating frequency could not be determined from a VSWR sweep; instead, a rough estimate of ≈ 1.5 GHz was obtained from the manually measured monopole length (≈ 5 cm) using the quarter-wave resonance relation. This is reasonably close to the 1.3 GHz actually used for the subsequent measurements.
- An initial Part B trial was corrupted by interference from an unrelated horn antenna belonging to another experiment, sitting only ≈ 1–2 m to the side of our setup; with it in place, received power fell (rather than rose) as the receiver approached the transmitter, a physically impossible trend that made the readings unusable until the interfering antenna was moved ≈ 3 m further away.
- After this fix, the received power increased as the transmitter–receiver distance decreased, in qualitative agreement with the expected inverse-square-law behaviour. However, the measured fall-off (slope ≈ −5.4, in units of 20 log10 R) was still considerably steeper than theory predicts; with only three data points spanning a narrow (≈ 1.5 m) range, this residual discrepancy is likely dominated by measurement/positioning uncertainty and by indoor multipath (reflections from the lab walls, floor, and furniture, and possibly some remaining stray coupling from the relocated antenna) rather than reflecting the antenna's true free-space behaviour.
- Both the co-polarized E-plane and H-plane beam patterns showed a clear main lobe near boresight (θ = 0°/360°) with several side lobes and no sharp nulls, consistent with a moderately directive but non-ideal dipole-horn combination measured at a limited number of angular samples (20° steps).
- Comparing the co-polarized and cross-polarized patterns, the cross-polarized readings were lower than the co-polarized ones at most angles in the E-plane, showing decent separation between the two polarizations. In the H-plane, the co- and cross-polarized readings were close to each other at many angles, showing weaker separation there.
- The measured half-power beamwidths were $\theta_E \approx 66.4°$ and $\theta_H \approx 41.2°$, giving an estimated directivity of D ≈ 15.1.
- The variation of the beam pattern with frequency was not investigated, as the co-/cross-polarization measurements were only carried out at a single frequency (1.3 GHz) due to time constraints, rather than being repeated at two additional frequencies as suggested in the manual.
- Principal sources of measurement uncertainty include interference from an unrelated nearby antenna during the initial Part B trial (mitigated, but possibly not fully eliminated, by relocating it ≈ 3 m away); the coarse (± 0.2–0.5 cm) manual measurement of the monopole length used to estimate the operating frequency; the small number (3) of distance points recorded in Part B; the 20° angular resolution of the beam-pattern sweeps, which limits the precision of the interpolated −3 dB points and hence the HPBW/directivity estimates; imperfect manual alignment of the antenna polarizations between co- and cross-polarization runs (especially for the H-plane cuts); and indoor multipath/reflections affecting all power readings.